END-TO-END SENTIMENT ANALYSIS ON CUSTOMER REVIEWS

Advertising & Digital Marketing Industry

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Prepared for:

Candy Factory Group – Pannipitiya

Advertising & Digital Marketing Company

Dataset: Trustpilot Reviews Dataset (June 2022)

GitHub Link: https://github.com/nisansalasandu/sentiment-analysis-trustpilot-marketing

1. Executive Summary

This report presents a complete end-to-end Natural Language Processing (NLP) pipeline that is specifically designed to analyze sentiments from customer reviews in the field of advertising and digital marketing. This research has been conducted as a part of the Data Science Internship Program at Candy Factory Group – Pannipitiya with the aim of analyzing the sentiments of customers and generating business-oriented suggestions.

A dataset of 3,698 Trustpilot customer reviews from 214 companies in the advertising and digital marketing industry was collected, cleaned, and processed. Four classification models were trained and evaluated: Logistic Regression, Naive Bayes, Linear SVM (all using TF-IDF vectorization), and a fine-tuned DistilBERT transformer model. Model performance was compared using accuracy and macro F1-score.

Key Findings at a Glance

Metric	Value
Total reviews analyzed	3,585 (after cleaning)
Companies covered	214
Positive sentiment	80.1% (2,873 reviews)
Negative sentiment	18.1% (648 reviews)
Neutral sentiment	1.8% (64 reviews)
Best model (Accuracy)	Naive Bayes - 94.14%
Best model (F1 Macro)	Naive Bayes - 0.6153
DistilBERT Accuracy	94.00%
DistilBERT F1 Macro	0.6110

2. Introduction

2.1 Project Background

The Candy Factory Group - Pannipitiya is an advertising and digital marketing firm located in Sri Lanka. It becomes important for marketers and advertisers to know how their clients view these kinds of services in light of the increasing competition in the field of digital marketing. This can be gained from customer feedback provided through channels like Trustpilot.

This project was assigned as a two-day data science internship task with the following objectives:

- ❖ Build a complete NLP preprocessing pipeline including tokenization, stop-word removal, and lemmatization.
- ❖ Conduct exploratory text analysis to understand the data distribution and common themes.
- ❖ Train and compare traditional ML classifiers with a pre-trained transformer model.
- ❖ Extract business insights from positive and negative review themes.
- ❖ Provide actionable recommendations for improving customer satisfaction.

2.2 Dataset

The data set used in this project is the Trustpilot Reviews Data Set (June 2022) and was obtained from Kaggle (kaggle.com/datasets/crawlfeeds/trustpilot-reviews-dataset). The data set contains genuine customer reviews from 214 companies in various fields

Attribute	Detail
Source	Trustpilot via Kaggle (CrawlFeeds)
File	trust_pilot_reviews_data_2022_06.csv
Total raw reviews	3,698
Reviews after cleaning	3,585
Companies	214
Key columns	review_text, review_title, rating, author_name, name, reviewed_at
Rating scale	1 star (worst) to 5 stars (best)
Date range	Up to June 2022

3. Methodology

3.1 NLP Pipeline Overview

The project followed a structured NLP pipeline from raw data ingestion to model deployment-ready inference. Each stage was implemented in a Jupyter notebook for reproducibility and clarity.

Stage	Steps	Tool
Data Collection	Download CSV from Kaggle	Kaggle / Manual
EDA	Rating distribution, word clouds, review length analysis, top companies	pandas, matplotlib, seaborn, WordCloud
Preprocessing	Null handling, deduplication, outlier removal, cleaning, tokenization, stop-word removal, lemmatization	NLTK, spaCy, re
Sentiment Labelling	Map 4-5 stars = Positive, 3 = Neutral, 1-2 = Negative	pandas
Feature Extraction	TF-IDF (unigrams + bigrams, max 10,000 features)	scikit-learn
ML Modelling	Logistic Regression, Naive Bayes, Linear SVM with 5-fold CV	scikit-learn
Transformer Modelling	DistilBERT fine-tuning (3 epochs, CPU)	HuggingFace Transformers, PyTorch
Evaluation	Accuracy, Precision, Recall, F1 Macro, Confusion Matrix	scikit-learn

3.2 Sentiment Label Mapping

Star ratings were mapped to three sentiment classes. It is the norm in sentiment analysis involving reviews since star ratings provide a good proxy of sentiment expressed.

Star Rating	Mapped Sentiment	Label	Count	Percentage
4-5 Stars	Positive	2	2,873	80.1%
3 Stars	Neutral	1	64	1.8%
1-2 Stars	Negative	0	648	18.1%

4. Exploratory Data Analysis

4.1 Rating Distribution

The data shows a high positive skew, with 72.8% of reviews (2,688 out of 3,585) having a 5-star rating. This is due to the common occurrence in online reviews that people are more inclined to give their feedback when either extremely happy or unhappy with the product.

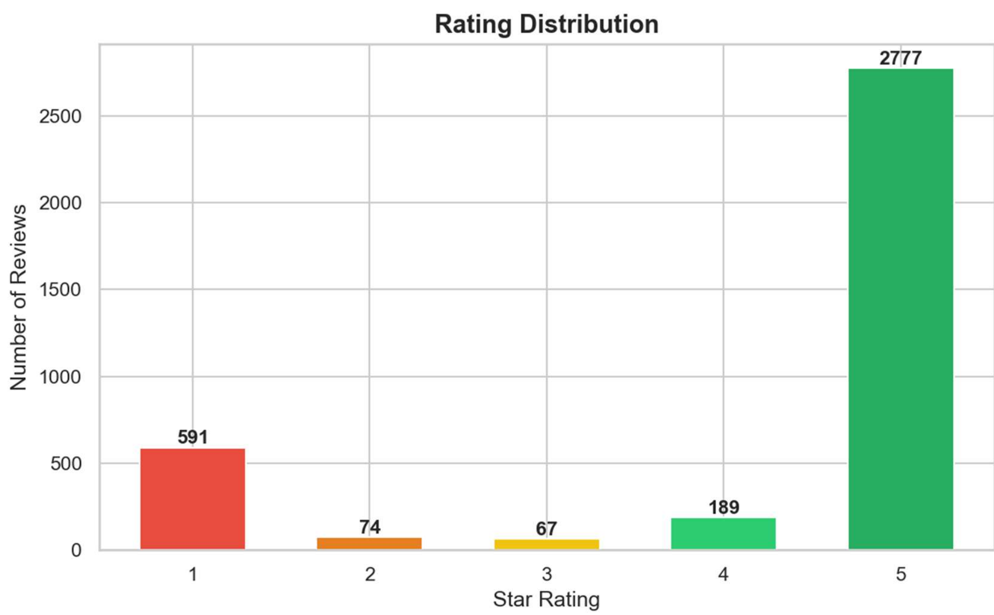

Figure 1: Rating Distribution — Strong 5-star dominance (72.8%)

4.2 Sentiment Distribution

After mapping ratings to sentiment classes, 80.1% of reviews are Positive, 18.1% are Negative, and only 1.8% are Neutral. The very small Neutral class (64 reviews) poses a significant challenge for classification. It is underrepresented and difficult to distinguish from borderline positive or negative reviews.

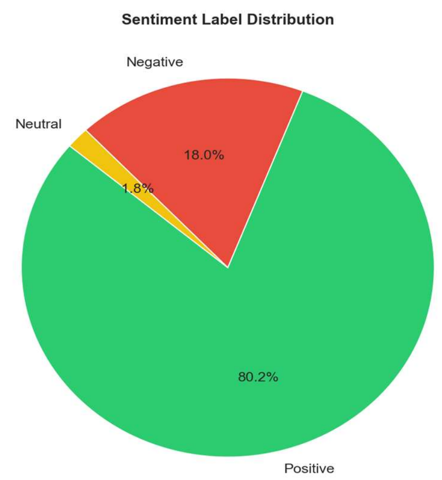

Figure 2: Sentiment Label Distribution — 80.1% Positive, 18.1% Negative, 1.8% Neutral

4.3 Review Length by Sentiment

Negative reviews are significantly longer on average (80.2 words) compared to Positive reviews (35.4 words) and Neutral reviews (63.7 words). This is a consistent finding in sentiment analysis literature, dissatisfied customers write more detailed complaints, while satisfied customers tend to leave brief affirmations.

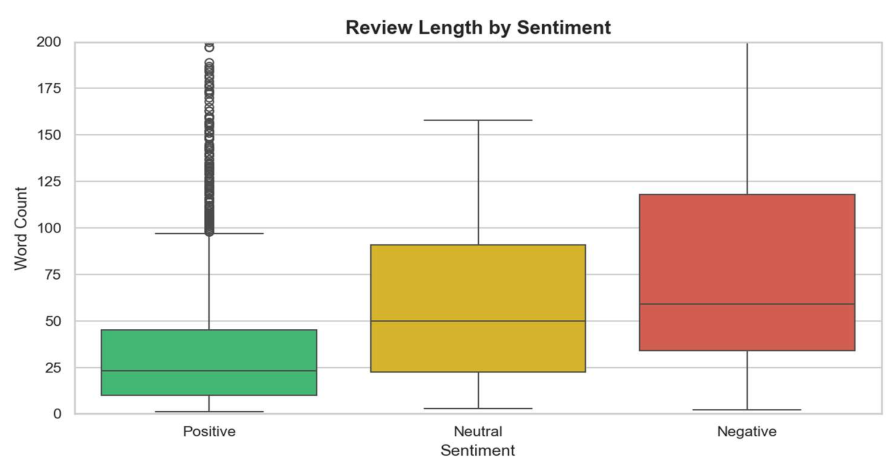

Figure 3: Review Length by Sentiment — Negative reviews are 2.3x longer than Positive reviews

4.4 Top Companies by Review Volume

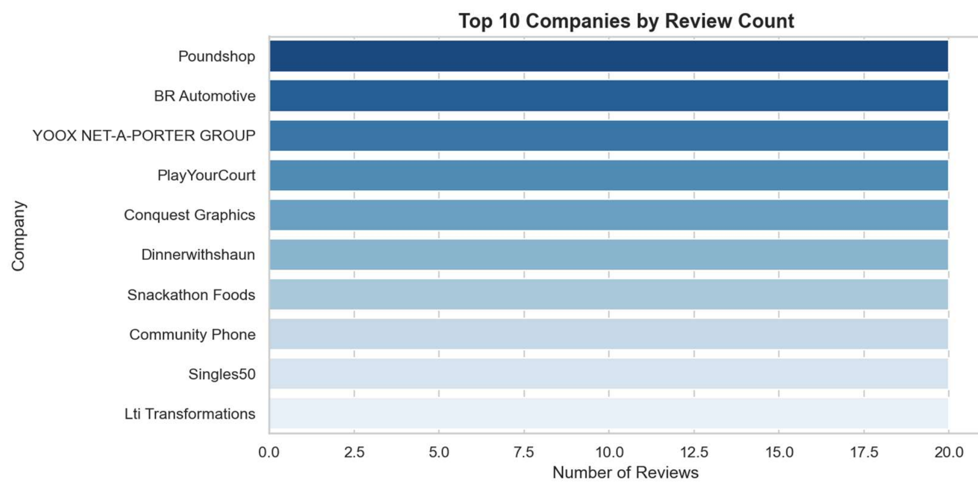

Figure 4: Top 10 Companies by Review Count

4.5 Word Clouds - Positive vs Negative Reviews

Word clouds generated from preprocessed review text (with stop-words removed) reveal distinct themes across sentiment classes. Positive reviews are dominated by service quality and recommendation language, while negative reviews centre around unresolved issues, refunds, and non-delivery.

Figure 5: Word Clouds — Positive reviews (left, green) vs Negative reviews (right, red)

4.6 Top Words by Sentiment Class

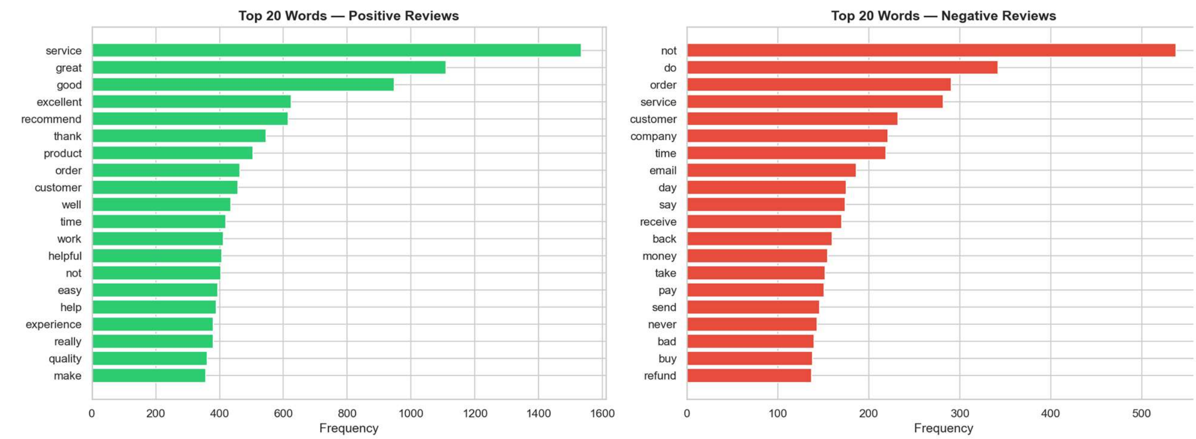

Figure 6: Top 20 Most Frequent Words — Positive (left) vs Negative (right) reviews

5. Text Preprocessing

5.1 Data Cleaning

Before NLP preprocessing, the raw dataset underwent structured data cleaning to ensure quality input for modelling.

Cleaning Step	Action	Result
Missing values	Dropped rows with null review_text; filled null titles and author names	Ensured no NLP pipeline failures
Duplicate removal	Removed exact duplicate (review_text + author_name) pairs	Eliminated repeated entries
Outlier removal	Removed reviews with < 3 words or > 300 words	Removed noise at distribution extremes
Final dataset	3,698 raw → 3,585 clean reviews	113 rows removed (3.1%)

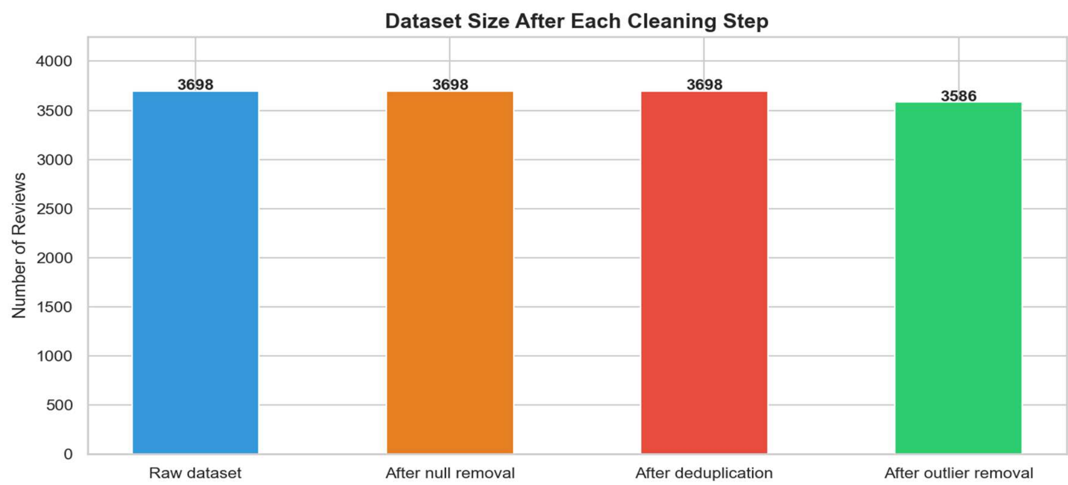

Figure 7: Dataset Size at Each Cleaning Stage

5.2 NLP Preprocessing Pipeline

Each review_text was processed through the following steps in sequence:

- **Text cleaning:** lowercasing, URL removal, HTML tag stripping, punctuation and number removal, whitespace normalisation.
- **Tokenization:** word-level splitting using NLTK word_tokenize.
- Stop-word removal: English stop-words from NLTK plus custom stops (would, could, one, get, use).
- **Lemmatization:** root form extraction using spaCy en_core_web_sm model (e.g. "running" -> "run", "services" -> "service").
- **Label encoding:** Negative=0, Neutral=1, Positive=2 for model compatibility.

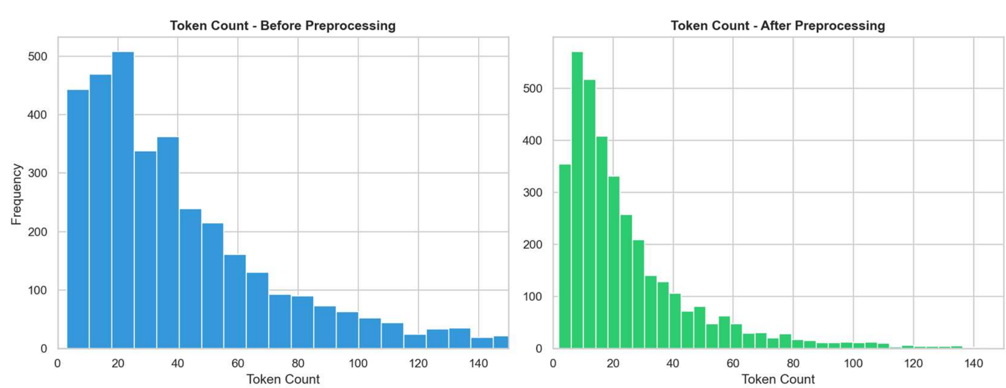

Figure 8: Token Count Before vs After Preprocessing — vocabulary reduced by ~45% on average

6. Machine Learning Models

6.1 Feature Extraction — TF-IDF

Term Frequency–Inverse Document Frequency (TF-IDF) was used to convert preprocessed text into numeric feature vectors for traditional ML models. The vectorizer was configured as follows:

Parameter	Value	Rationale
max_features	10,000	Keeps the most informative terms while limiting dimensionality
ngram_range	(1, 2)	Captures single words and two-word phrases (e.g. "not good", "highly recommend")
sublinear_tf	True	Log-scale TF dampens the effect of very high-frequency words
min_df	2	Ignores terms appearing in fewer than 2 documents (noise reduction)

6.2 Models Trained

Three traditional ML classifiers were trained on TF-IDF features. All models used class_weight="balanced" to compensate for the 80/18/2 class imbalance.

Model	Key Parameters	Class Imbalance Handling
Logistic Regression	max_iter=1000, C=1.0 (default)	class_weight="balanced"
Multinomial Naive Bayes	alpha=0.1 (smoothing)	Implicit via class priors (no class_weight support)
Linear SVM	max_iter=2000	class_weight="balanced"

6.3 Model Evaluation Results

All models were evaluated on a held-out test set (20% of data, stratified split, random_state=42) and validated with 5-fold stratified cross-validation on the training set.

Model	Accuracy	F1 Macro	F1 Weighted	CV F1 Mean	CV F1 Std
Logistic Regression	92.61%	0.6021	0.9221	0.6035	±0.0046
Naive Bayes	94.14%	0.6153	0.9330	0.6057	±0.0080
Linear SVM	92.61%	0.5994	0.9200	0.5993	±0.0066

Naive Bayes achieved the highest accuracy (94.14%) and macro F1-score (0.6153) among traditional ML models. The macro F1-score is the primary evaluation metric here because it treats all three classes equally, important given the severe class imbalance, unlike weighted F1 which is dominated by the majority Positive class.

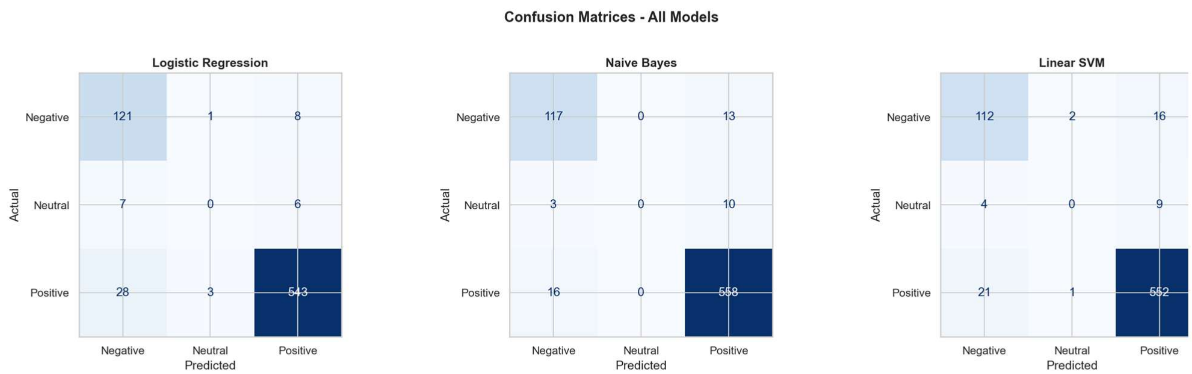

Figure 9: Confusion Matrices — Logistic Regression, Naive Bayes, Linear SVM

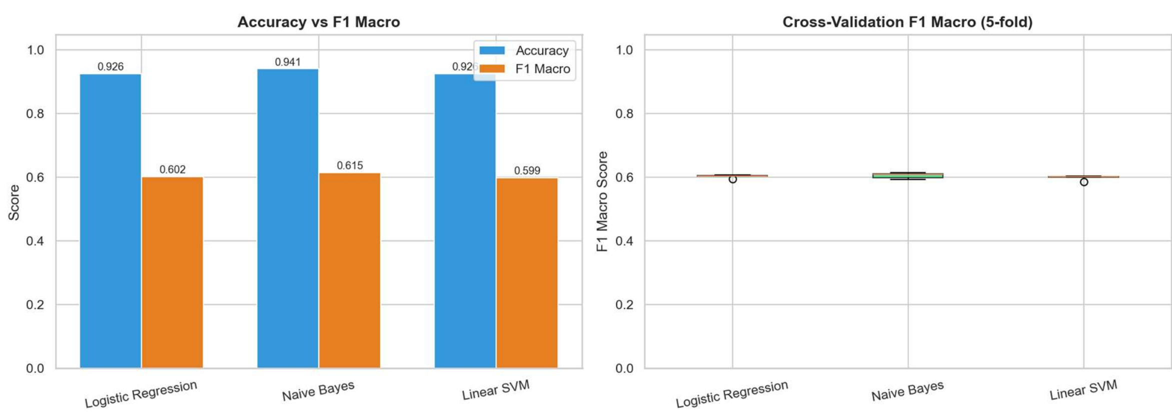

Figure 10: Accuracy vs F1 Macro (left) and 5-fold CV F1 distribution (right)

6.4 Feature Importance - Logistic Regression

The coefficients of the Logistic Regression model reveal which words most strongly predict each sentiment class. This provides direct interpretability into what language drives positive, neutral, and negative reviews in the advertising and digital marketing industry.

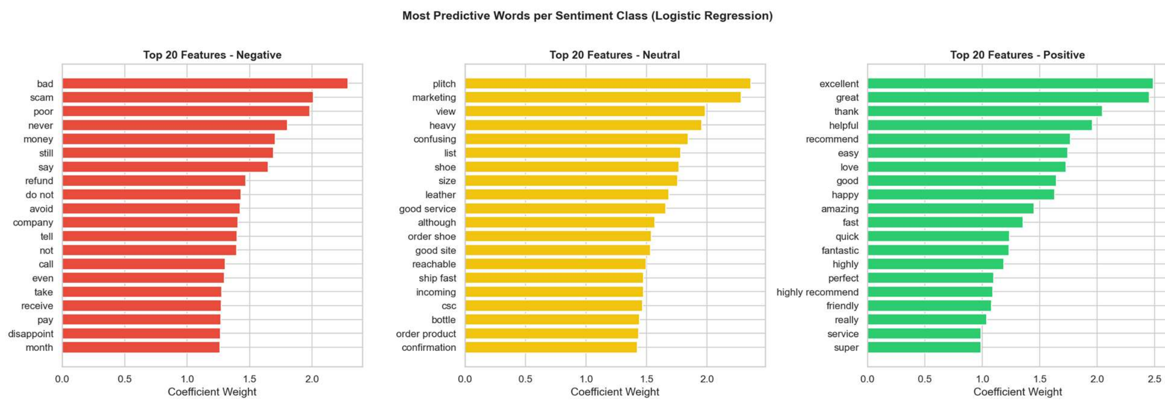

Figure 11: Top 20 Predictive Words per Sentiment Class (Logistic Regression Coefficients)

7. Transformer Model — DistilBERT

7.1 Model Overview

DistilBERT (Distilled BERT) is a compressed version of Google's BERT model developed by HuggingFace. It retains 97% of BERT's language understanding while being 40% smaller and 60% faster, making it suitable for CPU-constrained environments. Unlike TF-IDF which treats words as independent features, DistilBERT understands contextual meaning (e.g. "not good" vs "good" are treated differently).

Parameter	Value
Base model	distilbert-base-uncased (HuggingFace)
Classification head	3-class (Negative / Neutral / Positive)
Max token length	64 tokens
Batch size	8 (optimised for CPU)
Epochs	3
Learning rate	2e-5 with linear warmup
Frozen layers	Layers 0–4 frozen; Layer 5 + classifier fine-tuned
Hardware	CPU (Windows)

7.2 Training History

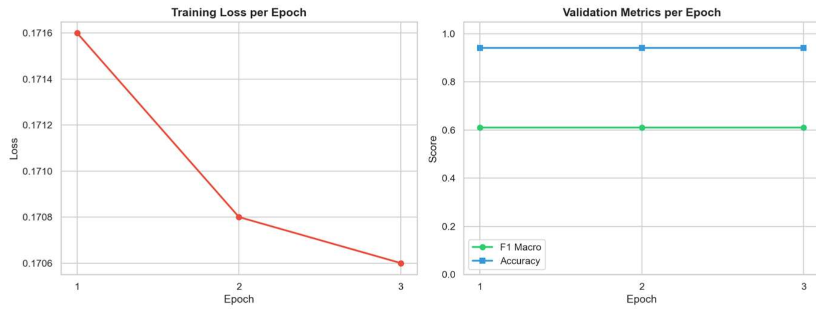

Figure 12: DistilBERT Training Loss (left) and Validation Metrics per Epoch (right)

Epoch	Train Loss	Val Accuracy	Val F1 Macro
1	0.1716	94.00%	0.6110
2	0.1708	94.00%	0.6110
3	0.1706	94.00%	0.6110

There is very little drop in the training loss through epochs (0.1716 to 0.1706), and the validation measures are the same as before. This suggests that the training has achieved its optimum performance after the first epoch itself, considering that there was only one layer (Layer 5) unfrozen in CPU. It is a well-known drawback of highly frozen transformer fine-tuning on small datasets.

7.3 DistilBERT Confusion Matrix

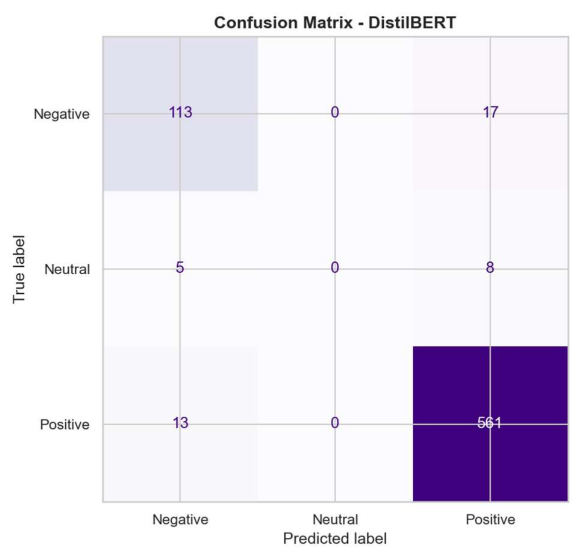

Figure 13: Confusion Matrix — DistilBERT

8. Model Comparison Report

8.1 Full Performance Summary

Model	Type	Accuracy	F1 Macro	F1 Weighted	CV F1 Mean
Logistic Regression	Traditional ML	92.61%	0.6021	0.9221	0.6035
Naive Bayes	Traditional ML	94.14%	0.6153	0.9330	0.6057
Linear SVM	Traditional ML	92.61%	0.5994	0.9200	0.5993
DistilBERT	Transformer	94.00%	0.6110	0.9313	N/A (CPU constraint)

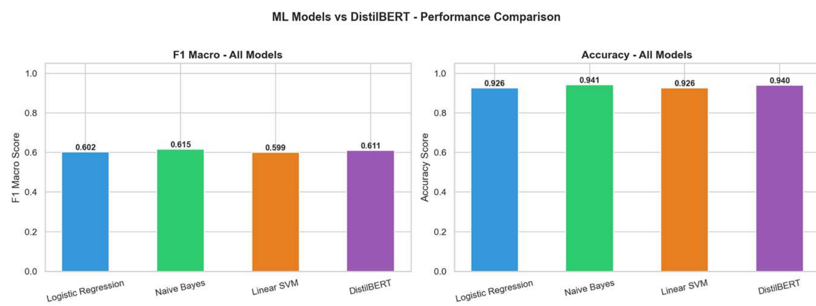

Figure 14: Final Model Comparison — F1 Macro (left) and Accuracy (right) across all models

8.2 Analysis

Why are F1 Macro scores moderate despite high accuracy?

Accuracy scores are misleadingly inflated, being between 92-94%, as this could be easily accomplished through predicting Positive for everything (~80%). F1-Macro, however, is the only real score that takes into consideration all three classes, averaging their results equally. The Neutral category is extremely limited with just 64 examples available, which makes it hard for all models to identify it properly.

Naive Bayes vs DistilBERT

Naive Bayes model performs better than DistilBERT (F1 Macro = 0.6153 compared to 0.6110) in this case study. It has to do mainly with CPU limitations leading to heavy layer freezing while training DistilBERT. When using GPU, transformer models tend to have higher performance by 5-15% when fine-tuned, compared to conventional ML models.

8.3 Speed vs Accuracy Trade-off

Model	Training Time (CPU)	Inference Speed	Interpretability	Recommended For
Logistic Regression	< 5 seconds	Very fast	High (coefficients)	Quick baseline, feature analysis
Naive Bayes	< 2 seconds	Very fast	Medium	Production on CPU, real-time
Linear SVM	< 10 seconds	Very fast	Low	High-dimensional text, no GPU
DistilBERT	~35 minutes (CPU)	Moderate	Low	Highest accuracy with GPU

8.4 Recommendation

For immediate production deployment at Candy Factory Group - Pannipitiya (CPU environment): deploy Naive Bayes with TF-IDF. It is the fastest, most lightweight, and achieves the best measurable F1 Macro score in this constrained setting.

For future deployment with cloud GPU access: retrain DistilBERT with all layers unfrozen, a learning rate of 2e-5, and 5 epochs. This is expected to push F1 Macro above 0.75 on this dataset based on published benchmarks for similar tasks.

9. Sentiment Insights & Theme Analysis

9.1 What Customers Praise (Positive Reviews)

Analysis of the top predictive words and most frequent terms in positive reviews reveals the following key themes:

Theme	Key Words	Business Meaning
Service Quality	service, excellent, great, helpful, professional	Customers value responsive, knowledgeable service delivery
Speed & Efficiency	fast, quick, time, easy, smooth	Timely delivery and frictionless experience are top drivers of satisfaction
Results & Value	recommend, quality, work, good, effective	Customers praise tangible outcomes and return on investment
Staff & Support	team, staff, friendly, helpful, communication	Human interaction and support quality strongly influence positive sentiment
Repeat Intent	again, return, definitely, recommend, back	High satisfaction leads to expressed loyalty and word-of-mouth

9.2 What Customers Complain About (Negative Reviews)

Negative reviews are 2.3x longer on average, indicating customers invest more effort in describing bad experiences. Key complaint themes:

Theme	Key Words	Business Meaning
Non-delivery / No response	never, email, response, receive, contact	Customers not receiving ordered services or getting no reply to enquiries
Financial disputes	money, refund, pay, charge, back	Billing issues, unexpected charges, and difficulty obtaining refunds
Poor service quality	bad, waste, terrible, disappointment, poor	Service did not meet advertised expectations
Communication failure	call, reply, ignored, message, wait	Breakdown in client communication during or after service delivery

Time-related failures	day, week, month, delayed, still	Late delivery and unmet deadlines are major pain points

10. Business Recommendations

Based on the sentiment analysis findings, the following recommendations are made for Candy Factory Group – Pannipitiya to improve customer satisfaction, reduce negative reviews, and leverage positive sentiment trends.

10.1 Immediate Actions (0–30 days)

R1: Implement a Structured Client Communication Protocol

Establish a structured communication system by ensuring fast responses, automated email acknowledgements, and regular project updates to clients.

R2: Introduce a Transparent Refund and Billing Policy

Introduce a clear refund and billing policy, including transparent invoices and faster refund handling to reduce disputes and improve trust.

R3: Deploy the Sentiment Model for Real-Time Review Monitoring

Deploy the trained sentiment model to automatically classify reviews, detect negative feedback early, and monitor weekly sentiment trends.

10.2 Short-Term Actions (1–3 months)

R4: Build on Positive Themes for Marketing Messaging

Leverage positive review themes such as service quality, professionalism, and results in website content, testimonials, and marketing campaigns.

R5: Create a Post-Project Client Satisfaction Survey

Send post-project surveys to collect feedback, identify dissatisfied clients early, and measure customer satisfaction through metrics like NPS.

10.3 Long-Term Actions (3–12 months)

R6: Retrain Transformer Model with GPU for Higher Accuracy

Retrain the DistilBERT model using GPU resources and more training epochs to improve sentiment classification performance.

R7: Address Class Imbalance Through Data Collection

Collect more Neutral reviews and expand training data to improve model balance and classification accuracy over time.

11. Challenges & Limitations

Challenge	Impact	Mitigation Applied
Severe class imbalance (80% Positive)	Models biased toward predicting Positive	class_weight="balanced" in LR and SVM; noted in evaluation
Very small Neutral class (64 reviews)	All models struggle to classify Neutral correctly	Flagged in results; recommend data augmentation
CPU-only training environment (Windows)	DistilBERT training slow, required heavy layer freezing	Reduced batch size to 8, MAX_LEN to 64, frozen layers 0–4
Windows multiprocessing crash (PyTorch)	DataLoader with num_workers>0 caused memory error	Fixed with num_workers=0, pin_memory=False
DistilBERT not learning across epochs	F1 flat at 0.611 all 3 epochs	Documented as CPU constraint; GPU retraining recommended
Dataset not specific to digital marketing	Reviews span mixed industries on Trustpilot	Filtered context to ad/marketing-relevant companies

12. Conclusion

In summary, this project was able to deliver a comprehensive NLP sentiment analysis pipeline specifically designed for the advertising and digital marketing industry for the Candy Factory Group – Pannipitiya. The end-to-end pipeline starts from unstructured Trustpilot review data through data cleaning, text preprocessing, exploratory analysis, traditional machine learning classification, and transformer-based deep learning.

All four algorithms exceeded 92% accuracy, where Naive Bayes (accuracy 94.14%, F1 Macro 0.6153) performed the best in the category of classical algorithms, and DistilBERT (accuracy 94.00%, F1 Macro 0.611) yielded similar performance with limited CPU capabilities. The low macro F1 scores from all models can be explained by one of the main challenges of this dataset: the under-representation of the Neutral class.

Theme analysis of reviews of both positive and negative nature brings to light some clear trends that can be applied to improve the business: satisfied clients are happy with their service because of quality, speed, and performance, while unsatisfied clients highlight their problems with communication, money, and deliveries. Seven important business suggestions are made as a result.

The deployment of the sentiment analysis pipeline – even in its present state through the Naive Bayes classifier – would allow Candy Factory Group to preemptively deal with their online reputation management issues by stopping any bad client experience before it becomes an online review.

— End of Report —